

Exam Date & Time: 13-May-2024 (01:45 PM - 04:45 PM)



MANIPAL UNIVERSITY JAIPUR

Even Semester End Term Examination, May 2024

Faculty of Science, School of Basic Sciences

Department of Computer Applications

DATA STRUCTURES AND ALGORITHMS [CA6206]

Marks: 80

Duration: 180 mins.

Section A

Answer all the questions.

All questions are compulsory.

(Memory Based Questions) (20-30 words Answers)

- 1) Differentiate between stack and queue. (2)
- 2) Explain different types of data structures. (2)
- 3) Define Doubly linked list. (2)
- 4) What is a graph? (2)
- 5) Define Recursion. (2)

Section B

Answer all the questions.

All questions are compulsory.

(Concept-Based Questions) (Maximum 150-200 Words Answers)

- 6) Solve the following recurrence relations using the master theorem. (a) $T(n) = 4T(n/2) + n^2$
(b) $T(n) = 2T(n/2) + n^2$ (6)
- 7) Explain the AVL tree and its properties. Construct an AVL tree with the following elements. Insert elements in the given order. Show all steps involved and label each step. (6)
4, 6, 5, 3, 1, 7, 10
- 8) Convert the following infix expression into a postfix expression using stack. Show every step in tabular form. (6)
 $A - (B / C + (D \% E * F) / G) * H$
- 9) Write an algorithm/C-Program to find the number of nodes in a given linked list. (6)

- 10) Construct a Binary Search Tree from the given values. 50, 70, 60, 20, 90, 10, 40, 100. Write down the preorder, post-order and inorder traversal of the tree constructed from the above given values. (6)

Section C

Answer all the questions.

All Questions are Compulsory.

(Analytical Based Questions) (Maximum 300-350 words Answers)

- 11) What do you understand by asymptotic notations? Describe types of asymptotic notations. (10)
- 12) What is meant by the Divide - and - Conquer approach? Write Divide - And - Conquer recursive Merge sort algorithm and derive the time complexity of this algorithm. Sort the following set of numbers using merge sort (Write step wise solution) 54, 26, 93, 17, 77, 31, 44, 55 (10)

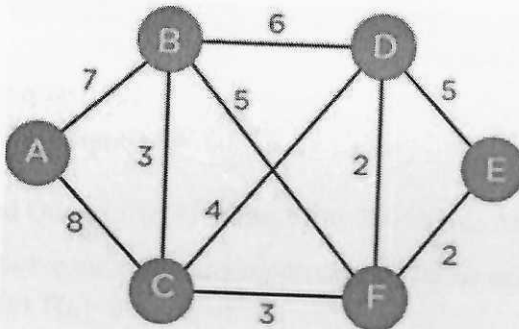
Section D

Answer all the questions.

All Questions are Compulsory.

(Case Study/Application-based Question) (Maximum 450-500 Words Answers)

- 13) Explain how the Matrix chain Multiplication problem can be solved using dynamic programming (DP). Find the minimum number of multiplication required to multiply the following matrices A1(10 x 6), A2 (6 x 8), A3 (8 x 12), and A4 (12 x 3) using DP (10)
- 14) Explain the greedy approach and find the minimum cost-spanning tree using the Kruskal algorithm from the graph shown below:-



(10)

-----End-----